

ESTAS-AQUABC Integration Guide

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1 Introduction

This document describes how the ESTAS-II transport framework and the AQUABC ecological model are coupled into a single executable. It covers the data flow at every interface boundary — from environmental forcing through pelagic kinetics to sediment diagenesis and back — and is intended for developers who need to modify, extend, or embed the coupled system.

1.1 Scope

Layer	Responsibility
ESTAS-II	Box-model topology, advection, dispersion, settling, mass loads, withdrawals, time integration, I/O
AQUABC Pelagic	Biogeochemical kinetics for 32 state variables (phytoplankton, zooplankton, nutrients, metals, redox, organic matter, dissolved oxygen, carbonate)
AQUABC Sediment	Multi-layer early diagenesis (24 state variables), bioturbation, bioirrigation, sediment–water fluxes
Auxiliary modules	Allelopathy (4 extra state variables), macroalgae (Droop quota), CO2SYS (carbonate equilibrium), exergy diagnostics

The integration follows a **caller–callee** pattern: ESTAS owns the time loop and calls into AQUABC subroutines at well-defined interface points.

2 Call Graph Overview

The top-level execution flow is:

```

ESTAS_II (main program)
|
+-- READ_AQUATIC_MODEL_INPUTS          [ESTAS]
|   +-- read INPUT.txt
|   +-- ALLOCATE_PELAGIC_ECOLOGY        [ESTAS/AQUABC bridge]
|   +-- INIT_WC_MODEL_CONSTANTS         [AQUABC]
|   +-- INIT_BSED_MODEL_CONSTANTS       [AQUABC/ESTAS bridge]
|   +-- read forcing time series
|
+-- RUN_SIMULATION                      [ESTAS]
|   |
|   +-- time loop (t = t_start ... t_end)
|       |
|       +-- SOLVE                      [ESTAS]
|           |
|           +-- UPDATE_TIME_FUNCS
|               |
|               +-- UPDATE_PELAGIC_DRIVING_FUNCS [bridge]
|               +-- interpolate flows, boundary concs
|               +-- interpolate settling velocities
|               +-- interpolate prescribed sediment fluxes
|           |
|           +-- CALC_DERIV
|               |
|               +-- compute transport derivatives
|               +-- PELAGIC_KINETICS          [AQUABC]
|               +-- (if Mode 2) coupling block:
|                   +-- FLX_ALUKAS_II_TO_SED_MOD_1_VEC [AQUABC]
|                   +-- AQUABC_SEDIMENT_MODEL_1        [AQUABC]
|                   +-- FLX_SED_MOD_1_TO_ALUKAS_II_VEC [AQUABC]
|               |
|               +-- Euler mass update
|               +-- write output
|
+-- cleanup / deallocate

```

3 Interface 1 — Driving Functions

ESTAS collects 10 environmental forcing values per box per time step and places them into an array that AQUABC reads as `DRIVING_FUNCTIONS(nkn, 10)`.

3.1 Driving Function Mapping

Index	Variable	Source	Units
1	Water temperature	Temperature time series	°C
2	Salinity	Salinity time series	PSU
3	Solar radiation (at box depth)	Solar radiation TS × Beer–Lambert attenuation through overlying boxes	W m ^{−2}
4	Fraction of day (photoperiod)	Fraction-of-day time series	dimensionless
5	Air temperature	Air temperature time series	°C
6	Wind speed	Wind speed time series	m s ^{−1}
7	Surface elevation	Computed from box volume	m
8	Water-column depth	V_i/A_i^s	m
9	Background light extinction	User-entered constant <code>K_B_E</code>	m ^{−1}
10	Ice cover fraction	Ice fraction time series	0–1

3.1.1 Light Attenuation Through Stacked Boxes

For vertically stacked boxes within a basin (box overlying another), solar radiation is attenuated using the saved light extinction coefficient from the box above:

$$I_{\text{box}} = I_{\text{surface}} \cdot \exp\left(-\sum_{k=1}^{n-1} K_{e,k} \cdot \Delta z_k\right)$$

where $K_{e,k}$ is the extinction coefficient stored in `SAVED_OUTPUTS(1)` of box k , and Δz_k is the depth of box k .

3.2 Flags

Five integer flags are passed per box:

Index	Meaning
1	Safe mode (always 0)

Index	Meaning
2	Surface box indicator (1 = surface, 0 = subsurface)
3	First time-step indicator (1 on first step, 0 thereafter)
4	Zooplankton option
5	Advanced redox simulation flag

4 Interface 2 — Pelagic Kinetics

4.1 The PELAGIC_KINETICS Call

At each time step, after computing all transport derivatives, ESTAS calls:

```
call PELAGIC_KINETICS(PELAGIC_BOX_MODEL_DATA, SEDIMENT_FLUXES, &
TIME, TIME_STEP, CALLED_BEFORE)
```

4.1.1 Inputs Passed to AQUABC

Array	Dimension	Description
STATE_VARIABLES(nkn, nstate)	$n_{kn} \times 32$	Current concentrations (g m^{-3})
DRIVING_FUNCTIONS(nkn, 10)	$n_{kn} \times 10$	Environmental forcing (see §3)
MODEL_CONSTANTS(nconst)	318	Pelagic kinetic and stoichiometric constants
FLAGS(nflags)	5	Control flags
SAVED_OUTPUTS(nkn, 5)	$n_{kn} \times 5$	Outputs carried forward from previous step
SEDIMENT_FLUXES(nkn, nstate)	$n_{kn} \times 32$	Prescribed sediment fluxes (Mode 1) or zero
TIME	scalar	Current simulation time (days)
TIME_STEP	scalar	Integration time step (days)
CALLED_BEFORE	integer	0 on first call, 1 thereafter

4.1.2 Outputs Returned to ESTAS

Array	Dimension	Description
DERIVATIVES(nkn, nstate)	$n_{kn} \times 32$	Kinetic reaction rates $R_{i,j}$ ($\text{g m}^{-3} \text{ day}^{-1}$)
PROCESS_RATES(nkn, nstate, NDIAGVAR)	$n_{kn} \times 32 \times 30$	Diagnostic process rates
SAVED_OUTPUTS(nkn, 5)	$n_{kn} \times 5$	Updated saved outputs (e.g. light extinction)
pH(nkn)	n_{kn}	Computed pH per box
CHLA(nkn)	n_{kn}	Chlorophyll-a per box (derived)

4.2 Integration of Kinetic Derivatives

The kinetic derivative is combined with sediment return fluxes and converted to a mass rate:

$$\left(\frac{dM_{i,j}}{dt} \right)_{\text{kin}} = (R_{i,j} + F_{i,j}^{\text{return}}) \cdot V_i$$

where $F_{i,j}^{\text{return}}$ is the sediment-to-water-column flux divided by depth (Mode 2) or zero.

5 Interface 3 — Settling and Deposition Fluxes

5.1 Settling Velocity and Dissolved Fractions

ESTAS interpolates per-box, per-variable settling velocities and dissolved fractions from time series and passes them as:

- `SETTLING_VELOCITIES(nkn, nstate)` — m day^{-1}
- `DISSOLVED_FRACTIONS(nkn, nstate)` — dimensionless, 0–1
- `FRACTION_OF_DEPOSITION(nkn, nstate)` — fraction deposited vs. reflected
- `DEPOSITION_AREA_RATIOS(nkn, nstate)` — bottom area correction factor

5.2 Water-Column to Sediment Flux Translation

The subroutine `FLX_ALUKAS_II_TO_SED_MOD_1_VEC` converts the 32 pelagic settling rates into 24 sediment deposition fluxes. This translation is necessary because the pelagic and sediment models use different state variable definitions.

5.2.1 Variable Mapping (WC → Sediment)

Sediment Variable	Index	Source WC Variables	Notes
NH ₄ -N	1	NH ₄ -N	Direct
NO ₃ -N	2	NO ₃ -N	Direct
PO ₄ -P (particulate organic N)	3	Det. Part. Org. N	Direct
Dissolved Org. N	4	Diss. Org. N	Direct (small)
PO ₄ -P	5	PO ₄ -P	Direct
Det. Part. Org. P	6	Det. Part. Org. P	Direct
Dissolved Org. P	7	Diss. Org. P	Direct (small)
Dissolved O ₂	8	Diss. O ₂	Direct
Det. Part. Org. C	9	Det. Part. Org. C + dead phytoplankton C	Aggregated
Dissolved Org. C	10	Diss. Org. C	Direct (small)
Dissolved Si	11	Diss. Si	Direct
Particulate Si	12	Part. Si + diatom Si	Aggregated
Inorganic C (DIC)	13	Inorg. C	Direct
Total Alkalinity	14	Tot. Alk.	Direct
Salinity	15	Driving function #2	From forcing
Fe(II), Fe(III), Mn(II), Mn(IV)	16–19	Metals	Direct
Ca, Mg, S(+6), S(-2), CH ₄	20–24	Minerals, sulphur, methane	Direct

The conversion handles the fundamental asymmetry: dead phytoplankton biomass (diatoms, cyanobacteria, other phytoplankton) in the water column is aggregated into detrital organic C, N, P in the sediment.

5.2.2 Settling Suppression

A chlorophyll-a-dependent factor suppresses settling velocities during intense blooms:

$$f_{\text{supp}} = f(\text{Chl-a})$$

This is computed inside `FLX_ALUKAS_II_TO_SED_MOD_1` before applying the settling flux calculation.

5.2.3 Not-Deposited Fluxes

Material that settles but is not deposited is “reflected” back to the water column:

$$F_j^{\text{not_dep}} = J_j^{\text{settl}} \cdot (1 - f_{\text{dep},j})$$

These fluxes are divided by water column depth and added to the kinetic derivative.

6 Interface 4 — Sediment Diagenesis (Mode 2)

When `MODEL_BOTTOM_SEDIMENTS = 2`, the full AQUABC sediment diagenesis model is called inside `CALC_DERIV`.

6.1 Pre-Call Setup

Before calling the sediment model, ESTAS performs:

1. **Temperature transfer:** sediment layer temperatures are set equal to water-column temperature (clamped to 0–45°C):

$$T_{\text{sed},\ell} = \max(0, \min(45, T_{\text{WC}}))$$

2. **Surface water concentration mapping:** 24 dissolved concentrations at the sediment–water interface are extracted from the current pelagic state:

Sediment Index	WC Source	Description
1	NH4_N	Ammonium nitrogen
2	NO3_N	Nitrate nitrogen
3	DET_PART_ORG_N	Particulate organic nitrogen
4	10^{-10}	Placeholder (dissolved org. N in sediment)
5	PO4_P	Phosphate phosphorus
6	DET_PART_ORG_P	Particulate organic phosphorus
7	10^{-10}	Placeholder (dissolved org. P in sediment)
8	DISS_OXYGEN	Dissolved oxygen
9	DET_PART_ORG_C	Particulate organic carbon
10	10^{-10}	Placeholder (dissolved org. C in sediment)
11	DISS_Si	Dissolved silica
12	10^{-10}	Placeholder (particulate Si in sediment)
13	INORG_C	Dissolved inorganic carbon
14	TOT_ALK	Total alkalinity
15	Driving function #2	Salinity
16–17	FE_II, FE_III	Iron species
18–19	MN_II, MN_IV	Manganese species
20–21	CA, MG	Calcium, magnesium
22–23	S_PLUS_6, S_MINUS_2	Sulphur species
24	CH4_C	Methane

Indices 4, 7, 10, 12 correspond to sediment dissolved/particulate organic pools that have no direct pelagic equivalent. They are initialised to a very small value (10^{-10}) to avoid division-by-zero errors inside the sediment model.

3. **Erosion/deposition thickness:** currently set to zero (`H_ERODEP = 0.0`), assuming negligible net erosion at the sediment surface.
4. **Settling flux conversion:** `FLX_ALUKAS_II_TO_SED_MOD_1_VEC` translates pelagic settling rates to 24-variable sediment deposition fluxes.

6.2 Sediment Model Call

```
call AQUABC_SEDIMENT_MODEL_1 &
  (nkn, INIT_SED_STATE_VARS, SED Depths, SED_POROSITIES, &
   SED_DENSITIES, PART_MIXING_COEFFS, SED_DIFFUSIONS, &
   SURF_MIXLEN, SED_BURRIALS, SURF_WATER_CONCS, SED_TEMPS, &
   NUM_SED_VARS, NUM_SED_LAYERS, SED_MODEL_CONSTANTS, &
   NUM_SED_CONSTS, SED_DRIVING_FUNCTIONS, NUM_SED_DRIV, &
   SED_FLAGS, NUM_SED_FLAGS, FLUXES_TO_SEDIMENTS, &
   NUM_FLUXES_TO_SEDIMENTS, NUM_FLUX_RECEIVING_SED_LAYERS, &
   ADVECTIVE_VELOCITY, TIME, TIME_STEP, H_ERODEP, &
   FINAL_SED_STATE_VARS, FLUXES_FROM_SEDIMENTS, &
   NUM_FLUXES_FROM_SEDIMENTS, PROCESSES_sed, NDIAGVAR_sed, &
   SED_OUTPUTS, NUM_SED_OUTPUTS, SED_SAVED_OUTPUTS, &
   NUM_SED_SAVED_OUTPUTS, SED_BURRIAL_RATE_OUTPUTS, &
   BOTTOM_SED_ADVANCED_REDOX_SIMULATION)
```

6.2.1 Key Input Arrays

Array	Shape	Description
INIT_SED_STATE_VARS	(nkn, layers, 24)	Initial sediment concentrations
SED_Depths	(nkn, layers)	Layer thicknesses (m)
SED_POROSITIES	(nkn, layers)	Volume fraction of pore water
SED_DENSITIES	(nkn, layers)	Bulk wet density (kg m^{-3})
PART_MIXING_COEFFS	(nkn, layers, 24)	Particle mixing / bioturbation ($\text{m}^2 \text{day}^{-1}$)
SED_DIFFUSIONS	(nkn, layers, 24)	Molecular + eddy diffusion ($\text{m}^2 \text{day}^{-1}$)
SED_BURRIALS	(nkn, layers)	Burial rates (m day^{-1})
FLUXES_TO_SEDIMENTS	(nkn, 24)	Deposition fluxes from settling ($\text{g m}^{-2} \text{day}^{-1}$)
SED_MODEL_CONSTANTS	(171)	Sediment reaction constants

6.2.2 Key Output Arrays

Array	Shape	Description
FINAL_SED_STATE_VARS	(nkn, layers, 24)	Updated concentrations
FLUXES_FROM_SEDIMENTS	(nkn, 30)	Return fluxes to water column
PROCESSES_sed	(nkn, 24, layers, 25)	Diagnostic process rates
SED_OUTPUTS	(nkn, layers, 26)	Additional outputs
SED_BURRIAL_RATE_OUTPUTS	(nkn, layers, 24)	Burial rate diagnostics

6.3 Post-Call Processing

After the sediment model returns:

1. **Negative clamping:** any negative sediment concentrations are set to zero.

2. **Flux back-translation:** FLX_SED_MOD_1_TO_ALUKAS_II_VEC maps the 30 sediment return fluxes back to the 32 pelagic state variables (30 mapped, 2 set to zero for living organisms that do not diffuse from sediment).
3. **Unit conversion:** return fluxes are divided by water-column depth to convert from $\text{g m}^{-2} \text{day}^{-1}$ to $\text{g m}^{-3} \text{day}^{-1}$:

$$F_{i,j}^{\text{return}} = \frac{F_j^{\text{sed} \rightarrow \text{WC}}}{z_i}$$

4. **State propagation:** FINAL_SED_STATE_VARS replaces INIT_SED_STATE_VARS for the next time step.

6.4 Sediment-to-Water Flux Mapping

The return mapping (FLX_SED_MOD_1_TO_ALUKAS_II) translates 30 sediment flux variables back to pelagic indices:

WC Flux Index	Sediment Flux Index	Variable
1	1	NH ₄ -N
2	2	NO ₃ -N
3	5	PO ₄ -P
4	8	Dissolved O ₂
5–8	—	Diatom C, Zooplankton C/N/P (zero — no living organism flux)
9–11	—	Detrital Part. Org. C/N/P (zero — remains in sediment)
12	9	Dissolved Org. C
13	3	Dissolved Org. N
14	6	Dissolved Org. P
15–16	—	Cyanobacteria, Other Phyto (zero)
17	11	Dissolved Si
18	12	Particulate Si
19	—	N ₂ -fixing Cyan. (zero)
20	13	Inorganic C (DIC)
21	14	Total Alkalinity
22–25	16–19	Fe(II), Fe(III), Mn(II), Mn(IV)
26–30	20–24	Ca, Mg, S(+6), S(-2), CH ₄

7 Interface 5 — Allelopathy

The allelopathy module adds 4 state variables (indices 33–36) appended after the standard 32. These are:

- Produced by phytoplankton groups proportional to their biomass
- Transported by the same advection–dispersion–settling equations
- Fed into `PELAGIC_KINETICS` which applies inhibition to susceptible groups

ESTAS handles allelopathy transparently: all transport arrays are dimensioned (`nkn`, `nstate + NUM_ALLOLOPATHY_STATE_VARS`), so the 4 extra variables ride the same transport infrastructure with no additional coupling code.

8 Interface 6 — Model Constants

8.1 Pelagic Constants

318 pelagic constants are initialised by `INIT_WC_MODEL_CONSTANTS` (AQUABC module) and stored in `MODEL_CONSTANTS(318)`. ESTAS can optionally override them from an external file specified as command-line argument 2 (`WCONST_*.txt`).

The override mechanism:

1. AQUABC fills defaults via `INIT_WC_MODEL_CONSTANTS`
2. If `USE_PELAGIC_CONSTANTS_FILE_NAME = 1`, ESTAS reads the override file and replaces matching entries

8.2 Sediment Constants

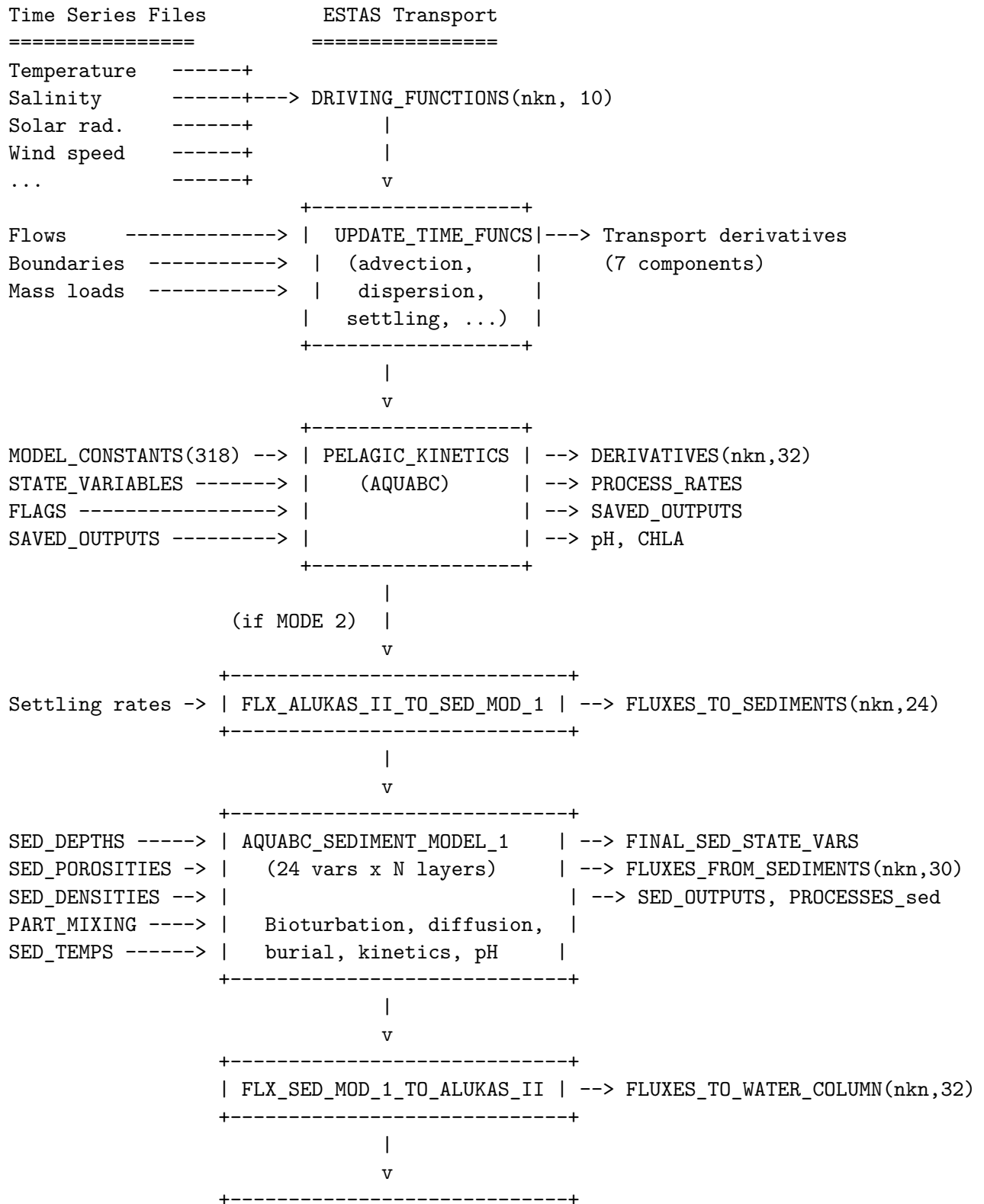
171 sediment constants are initialised by `INIT_BSED_MODEL_CONSTANTS` (in `mod_BOTTOM_SEDIMENTS.f90`) and optionally supplemented by `EXTRA_WCONST.txt`. The `isedi` parameter (required by the sediment model) is registered via SHYFEM's parameter system:

```
call para_insert_value('isedi', 0.0D0)
```

This ensures the sediment model can query `isedi` even when running in standalone ESTAS mode without SHYFEM.

9 Data Flow Diagram

The complete data flow per time step is:



```
|      EULER UPDATE      |
|  M(n+1) = M(n) + dM/dt*dt  |
|  C(n+1) = M(n+1) / V(n+1)  |
+-----+
```


10 Bioturbation Integration

The bioturbation module (`aquabc_II_sediment_bioturbation.f90`) was added in v0.2 and integrates at the sediment model level. Key integration points:

10.1 Within the Sediment Model

The sediment model calls eight bioturbation functions to compute depth- and temperature-dependent mixing coefficients:

Function	Purpose
<code>compute_bioturbation_Db</code>	Gaussian depth profile $D_b(z)$ scaled by temperature
<code>compute_Db_temperature_factor</code>	Q_{10} temperature correction
<code>compute_bioirrigation_alpha</code>	Exponential depth profile $\alpha(z)$ for bioirrigation
<code>compute_bioirrigation_enhancement</code>	Non-local exchange enhancement factor
<code>compute_dynamic_bulk_Db</code>	Combine baseline D_b with fauna-activity scaling
<code>compute_dynamic_dissolved_Db</code>	Pore-water component of bioturbation
<code>apply_lower_boundary_bio</code>	Ensure smooth decay at domain base
<code>compute_Db_at_depth</code>	Point evaluation for arbitrary depth

10.2 Particle Mixing Coefficients

The `PART_MIXING_COEFFS(nkn, layers, 24)` array, read from `BOTTOM_SEDIMENT_MODEL_INPUT.txt` and managed by ESTAS, provides baseline values. When bioturbation is active (`switch_partmixing > 0`), the sediment model dynamically overrides these with depth- and temperature-dependent values computed by the bioturbation module.

10.3 Bioirrigation

Bioirrigation enhances pore-water transport by multiplying the effective diffusion coefficient. This is handled entirely within the sediment model and is transparent to ESTAS.

11 Saved Outputs and State Propagation

11.1 Pelagic Saved Outputs

Five values are preserved across time steps per box:

Index	Variable	Usage
1	FE_II_DISS_OVER_FE_II	Iron(II) dissolved fraction
2	FE_III_DISS_OVER_FE_III	Iron(III) dissolved fraction
3	MN_II_DISS_OVER_MN_II	Manganese(II) dissolved fraction
4	MN_IV_DISS_OVER_MN_IV	Manganese(IV) dissolved fraction
5	DIP_OVER_IP	Dissolved inorganic P fraction

These are computed by PELAGIC_KINETICS and fed back on the next call. The light extinction coefficient, used for Beer–Lambert attenuation through stacked boxes, is also derived internally.

11.2 Sediment State Propagation

After each sediment model call:

$$\text{INIT_SED_STATE_VARS}^{n+1} = \text{FINAL_SED_STATE_VARS}^n$$

This ensures the sediment layers carry forward between time steps.

11.3 Called-Before Flag

The integer CALLED_BEFORE is 0 on the first invocation and 1 thereafter. AQUABC uses this to perform one-time initialisations (e.g. computing initial pH from alkalinity and DIC).

12 Output Integration

12.1 Pelagic Output

ESTAS writes per-box concentrations at `PRINT_INTERVAL` intervals. The output columns combine ESTAS transport information with AQUABC-computed values:

Columns	Content	Source
1	Time	ESTAS
2	Box number	ESTAS
3–34	32 pelagic state variables	AQUABC (via ESTAS transport)
35–38	4 allelopathic variables	AQUABC/Allelopathy

12.2 Sediment Output (Mode 2)

Three output files are written per time step:

1. **Sediment concentrations** — all 24 variables $\times$ all layers $\times$ all boxes
2. **Sediment fluxes to water column** — 30 flux variables per box (format: 36F20.10)
3. **Sediment burial rates** — optional COCOA-format output

12.3 COCOA Extended Output

When `PRODUCE_COCOA_OUTPUTS` > 0, additional files are produced:

- Pelagic process rates (all diagnostic variables)
- Sediment process rates per layer
- Bidirectional flux time series

13 Adding a New Coupled Variable

To add a new state variable that is transported by ESTAS and reacted by AQUABC:

13.1 Pelagic Side

1. Increment `nstate` in `mod_GLOBAL.f90`.
2. Add the variable index to `aquabc_pel_state_var_indexes.f90`.
3. Add kinetic equations in `aquabc_II_pelagic_kinetics.f90`.
4. Add initial conditions and any new constants.

13.2 ESTAS Side

5. Update settling velocity and dissolved fraction arrays (they auto-dimension from `nstate`).
6. Add the variable to output column mapping in `sub_WRITE_PELAGIC_OUTPUT.f90`.
7. If the variable interacts with sediments, update the flux translation routines:
 - `FLX_ALUKAS_II_TO_SED_MOD_1` (WC → sediment)
 - `FLX_SED_MOD_1_TO_ALUKAS_II` (sediment → WC)

13.3 Sediment Side (if applicable)

8. Increment `NUM_SED_VARS` in `mod_GLOBAL.f90`.
9. Add the variable to `BOTTOM_SEDIMENT_MODEL_INPUT.txt`.
10. Add kinetic reactions in the appropriate sediment library file.

14 Embedding in External Frameworks

ESTAS-AQUABC can be embedded in larger hydrodynamic frameworks (e.g. SHYFEM). The key interface contract:

Responsibility	Standalone ESTAS	Embedded in SHYFEM
Box topology	Read from files	Provided by FE mesh
Flows	Time series interpolation	Computed by hydrodynamics
Dispersion	Time series interpolation	Computed by turbulence model
Driving functions	Time series interpolation	From hydrodynamic solution
Time stepping	ESTAS Euler loop	SHYFEM time loop calls AQUABC
<code>isedi</code> parameter	Registered in <code>INIT_BSED_MODEL_CONSTANTS</code>	Registered by SHYFEM

The ecological model (AQUABC) is agnostic to the transport provider — it only sees `STATE_VARIABLES`, `DRIVING_FUNCTIONS`, `MODEL_CONSTANTS`, and `FLAGS`.

15 References

- AQUABC v0.2 Reference Manual (companion document)
- ESTAS-II Reference Manual (companion document)
- Boudreau, B.P. (1997). *Diagenetic Models and Their Implementation*. Springer.
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- Soetaert, K., Herman, P.M.J., and Middelburg, J.J. (1996). A model of early diagenetic processes. *Geochim. Cosmochim. Acta* 60, 1019–1040.